



University Of Information Technology And Sciences (UITs)

PROJECT REPORT

Project Title: House Price Estimation

Course: Simulation and Modeling Lab

Course Code: CSE-413

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Introduction:

In today's fast-paced and data-driven world, knowing the right price for a house can make a big difference — whether you're buying, selling, or investing. This project takes a practical approach to that challenge by using a linear regression model to estimate house prices based on real factors like the number of rooms, total area, and property location. By analyzing historical data and applying statistical techniques, the goal is to uncover patterns and relationships that help predict how much a house is likely to cost.

Objective:

- Build a predictive model using linear regression to estimate house prices.
- Clean and preprocess the dataset by handling missing values and converting non-numeric data into usable formats.
- Evaluate how well the model performs using both training and testing datasets.
- Implement everything in MATLAB, and generate results that are ready for submission or deployment.

Tools:

- Programming Language: MATLAB
- Libraries/Functions: Basic MATLAB built-in functions (readtable, fitlm, etc.)
- Datasets: train.csv, test.csv (provided in the project)
- Platform: MATLAB Simulation Environment

Methodology (Code Explanation):

1. Load and Prepare Data:

```
% Reading input files as tables
train_ref = readtable('train.csv',opts1);
test_ref = readtable('test.csv',opts2);
```

We start by loading the training and testing datasets into MATLAB using readtable, which reads tabular data directly from CSV files.

2. Handling Missing and Categorical Data:

To ensure our model isn't affected by missing or non-numeric values, we use a custom function called `replace_func.m`:

```
% data_set = replace_func(data_set,'Grv1',1,6);  
% data_set = replace_func(data_set,'Pave',2,6);  
|  
data_set = replace_func(data_set,'Reg',1,7);  
data_set = replace_func(data_set,'IR1',2,7);  
data_set = replace_func(data_set,'IR2',4,7);  
data_set = replace_func(data_set,'IR3',3,7);
```

This function cleans up the data by converting categorical values and filling in missing entries appropriately.

3. Training the Linear Regression Model:

```
% Next steps describe linear regression approach  
data_tr_tr_hat = [data_tr_tr ones(1250,1)];  
  
% Compute coefficients w and b that are included in w_hat  
% It is important to include 0.01*diag(ones(68)) to assure the  
% of the matrix does exist  
w_hat = (data_tr_tr_hat'*data_tr_tr_hat+0.01*diag(ones(68)))\(  
  
w = w_hat(1:67); % Extracting w  
b = w_hat(68); % Extracting b
```

We train a linear regression model using key features that strongly influence house prices, such as overall quality, living area, garage capacity, and basement size.

4. Making Predictions:

```
Y = w'*data_tr_te' + b; % Calculate the output values using li  
Y = [data(1251:1448,1) Y']; % Include "ID"s for the output  
  
challenge_output = w'*data_output' + b; % Calculate the output  
challenge_output = [data(1449:end,1) challenge_output']; % Inc
```

With the model trained, we use it to predict prices for the houses in the test dataset.

5. Exporting the Results:

```
challenge_output = ["Id" "SalePrice"; challenge_output];  
writematrix(challenge_output, 'output.csv')
```

Finally, we format the predictions into a table and save them as output.csv—a clean output ready for analysis or submission.

Results and Outcome:

The model successfully creates a file named submission.csv, containing predicted house prices for the test data. These predictions are based on relationships identified from the training data, and they reflect how the selected features impact housing prices in a realistic setting.

Conclusion:

Limitations:

- The model only uses a few features, which might cause it to miss other important factors influencing price.
- It assumes a linear relationship between inputs and outputs, which may not always reflect real market dynamics.
- There's no use of cross-validation or regularization, which could lead to overfitting or underfitting the data.

Future Improvements:

- Add feature selection techniques to automatically pick the most influential variables.
- Try more powerful algorithms like Ridge, Lasso, or Random Forests for better accuracy.
- Consider deploying the model in a web or mobile app for real-time predictions.
- Use cross-validation to ensure the model performs well on unseen data.

References:

- MATLAB Documentation: <https://www.mathworks.com/help/> Kaggle Housing Dataset (Ames):
- <https://www.kaggle.com/c/house-prices-advanced-regression-techniques>
- GitHub Repository: <https://github.com/MohammadShakibul/House-Price-Estimation>